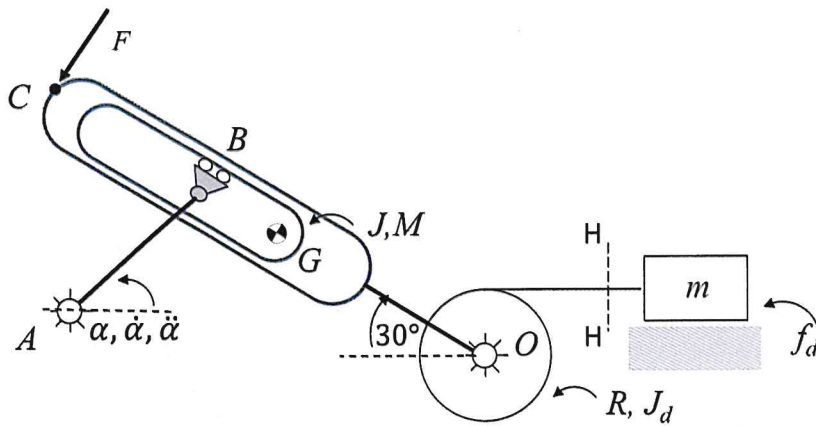
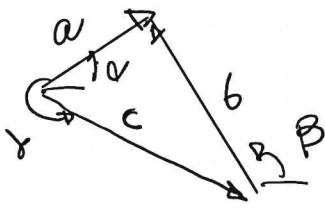




- 1) V_m, Q_m
- 2) F per garantire il moto
- 3) Tiro fune A-A



$$a e^{i\varphi} = b e^{i\beta} + c e^{i\gamma}$$

$$\begin{cases} a \cos \varphi = b \cos \beta + c \cos \gamma \\ a \sin \varphi = b \sin \beta + c \sin \gamma \end{cases}$$

$$\text{vel: } \dot{a} e^{i\varphi + \pi/2} = \dot{b} e^{i\beta + \pi/2} + \dot{c} e^{i\gamma}$$

$$\begin{cases} -\dot{a} \sin \varphi = -\dot{b} \sin \beta + \dot{c} \cos \beta \\ \dot{a} \cos \varphi = \dot{b} \cos \beta + \dot{c} \sin \beta \end{cases}$$

$$\begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} \begin{Bmatrix} \dot{b} \\ \dot{c} \end{Bmatrix} = \begin{Bmatrix} -\dot{a} \sin \varphi \\ \dot{a} \cos \varphi \end{Bmatrix}$$

$$\begin{cases} \dot{b} = -0,97 \\ \dot{c} = 0,129 \end{cases}$$

acc:

$$\ddot{a} e^{i\varphi + \pi/2} - \dot{a}^2 e^{i\varphi} = \ddot{b} e^{i\beta} + \dot{b} \dot{\beta} e^{i\beta + \pi/2} + 2\dot{b} \dot{c} e^{i\beta + \pi/2} - \dot{b}^2 e^{i\beta}$$

$$\begin{cases} -\ddot{a} \sin \varphi - \dot{a}^2 \cos \varphi = \ddot{b} \cos \beta - \dot{b} \dot{\beta} \sin \beta - 2\dot{b} \dot{c} \sin \beta - \dot{b}^2 \cos \beta \\ \ddot{a} \cos \varphi - \dot{a}^2 \sin \varphi = \ddot{b} \sin \beta + \dot{b} \dot{\beta} \cos \beta + 2\dot{b} \dot{c} \cos \beta - \dot{b}^2 \sin \beta \end{cases}$$

$$\begin{bmatrix} \cos \beta & -b \sin \beta \\ \sin \beta & b \cos \beta \end{bmatrix} \begin{Bmatrix} \ddot{b} \\ \ddot{\beta} \end{Bmatrix} = \begin{Bmatrix} -\ddot{\alpha} a \sin \psi - a \dot{\psi}^2 \cos \psi + 2b \dot{\beta} \dot{\psi} \sin \beta + \dot{\beta}^2 b \cos \beta \\ \ddot{\alpha} a \cos \psi - a \dot{\psi}^2 \sin \psi - 2b \dot{\beta} \dot{\psi} \cos \beta + \dot{\beta}^2 b \sin \beta \end{Bmatrix}$$

$$\begin{cases} \ddot{b} = -0,67 \\ \ddot{\beta} = 0,74 \end{cases}$$

$$\begin{cases} \vec{v}_m = -R \dot{\beta} \vec{\tau} = -0,13 \\ \vec{a}_m = -R \ddot{\beta} \vec{\tau} = -0,74 \end{cases}$$

$$(2) \frac{\Delta E_c}{\Delta t} = \sum W$$

$$E_c = \frac{1}{2} M V_a^2 + \frac{1}{2} J \omega^2 + \frac{1}{2} m V_m^2$$

$$\frac{\Delta E_c}{\Delta t} = M \vec{v}_a \cdot \vec{a}_a + (J) \vec{\omega} \cdot \vec{\dot{\omega}} + m \vec{v}_m \cdot \vec{a}_m$$

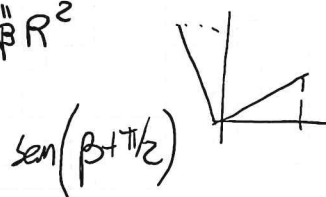
$$M \dot{\beta} \ddot{\beta} d^2 + (J) \dot{\beta} \ddot{\beta} + m \dot{\beta} \ddot{\beta} R^2$$

$$\sum W = M \vec{g} \cdot \vec{v}_a + \vec{F} \cdot \vec{\omega} + \vec{T} \cdot \vec{v}_m$$

$$\begin{aligned} & M g \dot{\beta} d \sin(\beta + \pi/2) + F \cdot \dot{\beta} l - T \cdot R \dot{\beta} \\ & = + M g \dot{\beta} d \cos \beta + F \dot{\beta} l - l M g R / \dot{\beta} \end{aligned}$$

$$F = \frac{(M d^2 + (J) + m R^2) \ddot{\beta} - M g d \cos \beta \dot{\beta} + l M g R / \dot{\beta}}{\dot{\beta} l}$$

11,66



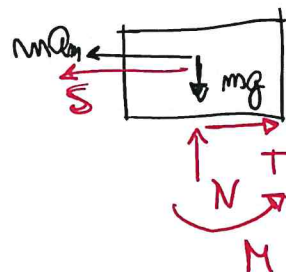
$$\sum F_y = 0$$

$$N = mg$$

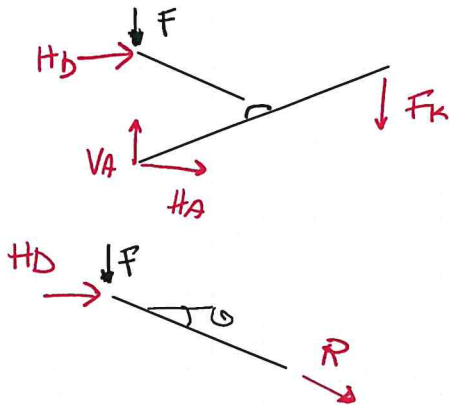
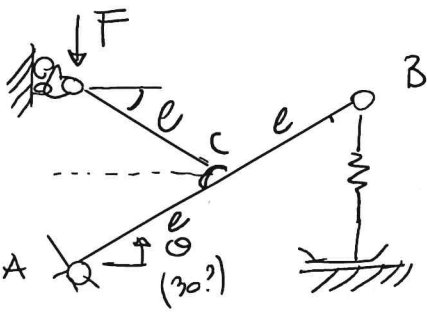
$$\sum F_x = 0$$

$$S = -m a_m + l M g$$

$$S = 56,4$$



- 1) R.V. in A B, C, D
- 2) $\varnothing$ e K noto trovare l_0
- 3) Azioni interne AB



$$\begin{aligned} \sum F_x &= 0 & H_B + H_A &= 0 & H_A &= -F\sqrt{3} \\ \sum F_y &= 0 & V_A &= F + F_K & V_A &= 0 \\ \sum M_A &= 0 & -H_B 2l \sin 30^\circ - F_K 2l \cos 30^\circ &= 0 \\ & \Rightarrow F_K &= -H_B \tan 30^\circ & F_K &= -F \end{aligned}$$

$$\begin{aligned} \sum M_C &= 0 & -H_B l \sin 30^\circ + F l \cos 30^\circ &= 0 & H_B &= F\sqrt{3} \\ \sum F_y &= 0 & -F - R \sin 30^\circ &= 0 & R &= 2F \end{aligned}$$



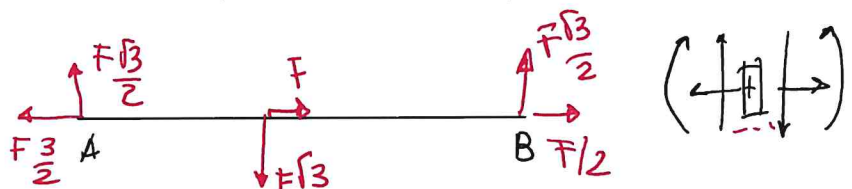
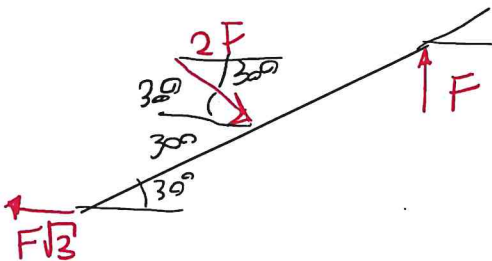
$$F_K = K \Delta l = k(L - l_0) = k(2l \sin 30^\circ - l_0)$$

$$-F = k\left(\frac{2l}{2} - l_0\right) \quad \text{con } k = F/l$$

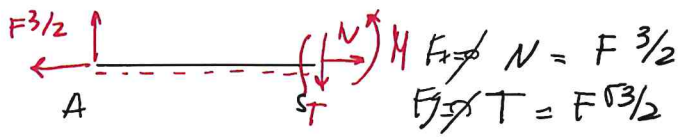
$$-F = \frac{F}{l}(l - l_0)$$

$$2l = l_0$$

$$l_0 = 2l$$



$$F\sqrt{3}/2$$

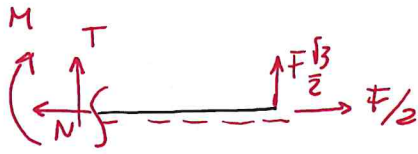


$$F_x = F \cos 30^\circ = F \frac{\sqrt{3}}{2} \quad N = F \frac{3}{2}$$

$$F_y = F \sin 30^\circ = F \frac{1}{2} \quad T = F \frac{\sqrt{3}}{2}$$

$$\sum M_s = 0 \quad M(x) = F \frac{\sqrt{3}}{2} x \quad M(0) = 0$$

$$M(l) = F \frac{\sqrt{3}}{2} l$$

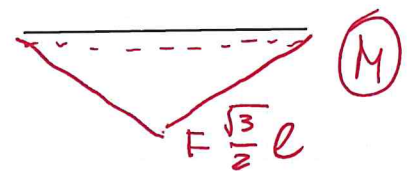
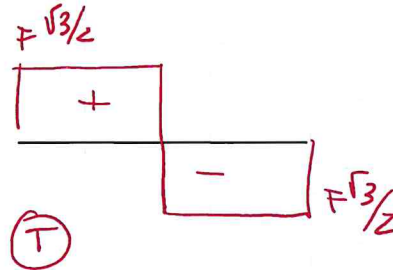
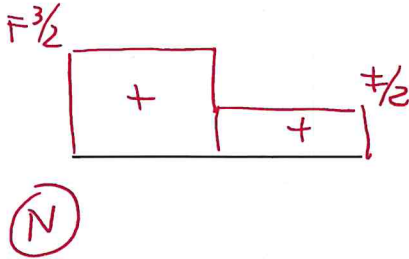


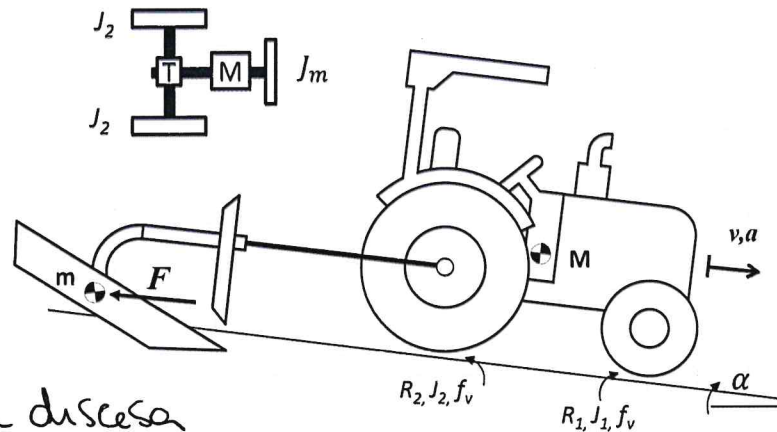
$$F_x = 0 \quad N = F/2$$

$$F_y = F \sin 30^\circ = F/2 \quad T = -F \frac{\sqrt{3}}{2}$$

$$\sum M_s = 0 \quad M(x) = F \frac{\sqrt{3}}{2} x \quad M(0) = 0$$

$$M(l) = F \frac{\sqrt{3}}{2} l$$





- 1) Regime in discesa
- 2) Acc discesa con $C_m = XX$ applicato
- 3) Regime discesa staccando l'altro

$$① W_m = C_m \omega_m$$

$$W_u = M \bar{g} \bar{V} + M_q \bar{g} \bar{V} + \bar{F} \bar{V} - 2(N_A + N_P) f_v |V|$$

$$2(N_A + N_P) = Mg \cos \alpha$$

$$\omega_R = \gamma \omega_m$$

$$V = \gamma \omega_m R_2$$

$$W_u = (M + M_q) g \sin \alpha \gamma \omega_m R_2 - F \sin \alpha \gamma \omega_m R_2 - Mg \cos \alpha f_v R_2 / \omega_m / \gamma$$

$$W_u = \underbrace{\gamma R_2 [(M + M_q) g \sin \alpha - F \sin \alpha - Mg \cos \alpha f_v]}_{C_k < 0} \omega_m$$

modo diretto

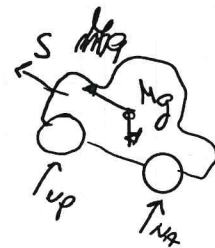
$$\eta_0 W_1 + W_2 = 0$$

$$\eta_0 C_m \omega_m + \gamma \omega_m R_2 [(M + M_q) g \sin \alpha - F \sin \alpha - Mg \cos \alpha f_v] = 0$$

①

$$C_{mR} = - \frac{1}{\eta_0} C_k$$

54,7



$$C_{mR} = -\frac{1}{\eta_D} C_{m\dot{\theta}} \quad 54,7$$

$$W_1 = \overbrace{C_{m\dot{\theta}} \dot{\theta}}^{>0} - \underbrace{J_m \dot{\theta} \ddot{\theta}}_{?}$$

$$W_2 = W_U - 2J_2 \omega_2 \dot{\omega}_2 - 2J_1 \omega_1 \dot{\omega}_1 - (M+M_a) v a$$

$$W_U = \left(2J_2 \gamma^2 \dot{\omega}_m \dot{\omega}_m + 2J_1 \left(\gamma \frac{R_2}{R_1} \right)^2 \dot{\omega}_m \dot{\omega}_m + (M+M_a) \gamma^2 \dot{\omega}_m \dot{\omega}_m R_2^2 \right)$$

$$W_U = \underbrace{\gamma^2 \dot{\omega}_m \dot{\omega}_m \left(2J_2 + 2J_1 \left(\frac{R_2}{R_1} \right)^2 + (M+M_a) R_2^2 \right)}_{>0}$$

con $\overline{C_{m\dot{\theta}}} > C_{mR}$ Hp: $a > 0 \Rightarrow$ moto diretta

$$\eta_D W_1 + W_2 = 0 \quad W_2 = W_U - \dot{\omega}_m \dot{\omega}_m \underbrace{\gamma^2 \left(2J_2 + 2J_1 \left(\frac{R_2}{R_1} \right)^2 + (M+M_a) R_2^2 \right)}_{23,9}$$

$$(\eta_D \overline{C_{m\dot{\theta}}} - \frac{1}{2} J_m \dot{\omega}_m) \dot{\omega}_m + W_U - \dot{\omega}_m \dot{\omega}_m J^* = 0 \quad J^* \quad 23,9$$

$$(\eta_D J_m + J^*) \dot{\omega}_m \cancel{\omega_m} = \left[\eta_D \overline{C_{m\dot{\theta}}} + \gamma R_2 ((M+M_a) g \sin \varphi - F_{\text{sent}} - M g \cos \varphi) \right] \cancel{\omega_m}$$

$$\dot{\omega}_m = \frac{1}{\eta_D J_m + J^*} \left[\eta_D \overline{C_{m\dot{\theta}}} + \gamma R_2 ((M+M_a) g \sin \varphi - F_{\text{sent}} - M g \cos \varphi) \right] = 4,83$$

(2)

$$W_1 = C_{m\dot{\theta}} \dot{\theta}$$

$$W_2 = \gamma \dot{\omega}_m R_2 \underbrace{[M g \sin \varphi - M g \cos \varphi]}_{>0}$$

>0 moto Retragrada 65,9

$$W_1 + \eta_R W_2 = 0$$

$$C_{m\dot{\theta}} \cancel{\dot{\theta}} + \eta_R \gamma \cancel{\dot{\omega}_m} R_2 [M g \sin \varphi - M g \cos \varphi] = 0$$

$$C_{m\dot{\theta}} = -\eta_R \gamma R_2 [M g \sin \varphi - M g \cos \varphi] = -46,16$$